# Local Epochs, Aggregation and Variable Selection in Federated Coordinate-Descent Lasso

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**Abstract**

Changing local epochs need not repair the variable-selection defects of primal averaging in federated Lasso. We diagnose this issue for cyclic coordinate descent, separating aggregation, penalty selection, and cost. Under orthogonal local designs, one sweep determines every local fit, making the averaged estimator independent of epoch count; its support generally inherits the union of local supports. Simulations with 200 replicates per design show that local penalty selection changes support recovery. Post-fit thresholding provides an inexpensive retrofit, whereas per-round thresholding requires multiple candidate trajectories. In a separate common-penalty comparison, adapted FedDualAvg gives substantially smaller objective gaps and better support recovery than CD averaging at equal selected-path uploads, with threefold training-path cost for learning-rate selection. These results favour an established composite solver when pooled-objective accuracy is required. The contribution is a diagnosis of CD averaging and its practical retrofits, with explicit limits on comparisons across tuning rules, communication budgets, and local computations.

**Keywords:** federated learning; Lasso; coordinate descent; variable selection; communication cost; dual averaging.

## 1 Introduction

Multi-site statistical analysis is increasingly constrained by data that cannot be centralised: electronic health records behind institutional firewalls, trials governed by separate data-use agreements, claims databases that no single sponsor controls. Federated learning (McMahan et al., 2017; Kairouz et al., 2021) addresses this by exchanging model parameters rather than records. A central server broadcasts a current parameter vector, each site updates it on its own data, and the server aggregates the returned vectors.

Let $E$ denote the number of local passes—here, full coordinate-descent (CD) epochs—each site performs between aggregation rounds. Larger $E$ can reduce communication, but also allows local iterates to move further toward different site-specific optima, producing client drift (Li et al., 2020b; Karimireddy et al., 2020). For the Lasso (Tibshirani, 1996), both local fitting and server aggregation can affect which coefficients are exactly zero. Consequently, objective values, prediction error and agreement of selected variables can respond differently to a change in $E$.

This article studies the choice of $E$ for a synchronous federated coordinate-descent Lasso in which each site tunes its own penalty on a local validation split and the server aggregates by sample-size weight. The question is how to interpret changes in its estimated active set as $E$ varies.

Such changes can reflect local optimisation, averaging of different sparse vectors, penalty selection and the numerical threshold used to declare a coefficient active. We examine these components through an orthogonal calculation and controlled computational comparisons.

The central conclusion is that changing local epochs is insufficient to resolve the aggregation problem. In the common-penalty experiment, adapted FedDualAvg achieves much smaller objective gaps and higher support-recovery $F_1$ than CD averaging at the displayed training budgets. This supports using an established composite solver when the aim is to solve the pooled penalised objective. Our thresholding variants instead evaluate modifications to an existing CD-averaging workflow: post-fit ST adds a small validation exchange, while per-round P requires a separate fit for each threshold. Their role is to characterise practical CD modifications under the tuning procedures and costs specified below.

**Contributions.** Our contributions are as follows.

1. **An exact orthogonal calculation.** Under orthogonal local designs, the CD iteration reaches $\sum_j w_j \mathcal{S}(z_j, \lambda_j)$ after one round, independently of $E$ (Proposition 2). Except for exact cancellations, its support is the union of the local supports and contains the pooled Lasso support. The calculation identifies exactly why the epoch count has no effect in this setting.

2. **False positives under a fixed penalty.** Corollary 1 gives an explicit null-selection probability under independent Gaussian site statistics and a fixed common penalty. For balanced partitions of a fixed sample, this probability strictly increases with the number of sites.

3. **A common objective for comparing solvers.** Sample-size weights make the weighted local objective equal to the pooled Lasso objective at penalty $\bar{\lambda}$ (Lemma 1). The averaged fixed-point condition need not imply stationarity of that objective (Proposition 3). Consensus ADMM (Boyd et al., 2011) supplies an optimisation benchmark at the same penalty, separating objective accuracy from support recovery.

4. **Separating tuning from aggregation.** We compare server thresholding (FedAvg-ST), per-round proximal aggregation (FedAvg-P), and consensus ADMM with pooled and local baselines. Minimum-validation-error and one-standard-error variants expose the effect of the selection rule. A communication-budget comparison with federated dual averaging shows that an established composite method substantially reduces the objective error of common-penalty CD averaging. The comparison separates optimisation accuracy from the practical thresholding study.

5. **Communication and local work are separate costs.** We report both transmitted bytes and local computation, including additional federated training paths required for tuning. Increasing $E$ can save communication while increasing total local work, so the two measures answer different deployment questions.

**Related work.** The local-epoch trade-off is well studied for smooth objectives through analyses of local SGD (Stich, 2019; Li et al., 2020b) and drift corrections such as FedProx (Li et al., 2020a) and SCAFFOLD (Karimireddy et al., 2020). Composite and proximal federated methods address nonsmooth regularisation (Tran Dinh et al., 2021). In the statistics literature, communication-efficient distributed sparse estimation includes one-shot and surrogate-likelihood constructions (Lee et al., 2017; Battey et al., 2018; Jordan et al., 2019) and distributed optimisation with consensus (Boyd et al., 2011; Duchi et al., 2012; Wang et al., 2017).

Closest to this article is the federated composite optimisation literature. Yuan et al. (2021) identify the "curse of primal averaging", introduce FedDualAvg, and evaluate sparse recovery using precision, recall, sparsity density and $F_1$. Bao et al. (2022) develop faster dual-averaging methods and statistical recovery guarantees, and explicitly report support-recovery $F_1$ for sparse linear regression. Thus both the averaging mechanism and variable selection as an evaluation target have established precedents.

Our contribution combines an explicit calculation for local cyclic coordinate descent, a fixed-penalty null-selection formula, and a computational assessment of local epochs under stated tuning and cost conventions. The analysis complements established support-recovery and stability-selection work (Zhao and Yu, 2006; Meinshausen and Bühlmann, 2010). The simulations examine the behaviour of this particular protocol beyond its solvable orthogonal case. A fixed synthetic benchmark provides an additional illustration of agreement between fitted supports; its unknown generating support precludes a recovery assessment.

The remainder of the article is organised as follows. Section 2 fixes notation and states the algorithm. Section 3 contains the analytical results. Section 4 introduces FedAvg-ST and the comparators. Sections 5 and 6 describe and report the Monte Carlo study, Section 7 compares fixed selected-path upload budgets, Section 8 presents the fixed synthetic benchmark, and Section 9 discusses limitations.

# 2 Setup and algorithm

## 2.1 Notation

There are $K$ sites. Site $j$ holds $X_j \in \mathbb{R}^{m_j \times p}$ and $y_j \in \mathbb{R}^{m_j}$, with $m_j > 0$, $m = \sum_{j=1}^{K} m_j$ and aggregation weights $w_j = m_j/m$. Assume that every local predictor column has positive squared norm, so the coordinate updates below are well defined. Write $X = (X_1^\top, \ldots, X_K^\top)^\top$ and $y$ for the concatenated data. This notation defines a benchmark; the federated updates do not require the records to be pooled. The local penalised objective at site $j$ is

$$\mathcal{L}_j(\beta) = \frac{1}{2m_j} \|y_j - X_j\beta\|_2^2 + \lambda_j \|\beta\|_1, \qquad \beta \in \mathbb{R}^p, \tag{1}$$

with a site-specific penalty $\lambda_j > 0$ selected once, before federated fitting, on the site's own validation split, and then held fixed. We compare minimum-validation-error and one-standard-error selection rules, as specified in Section 5. The scale $1/(2m_j)$ removes the direct dependence of the loss on sample size; predictor and response scales still affect the appropriate penalty. The common objective used for the optimisation comparison is

$$F(\beta) = \sum_{j=1}^{K} w_j \mathcal{L}_j(\beta). \tag{2}$$

For a vector $v$ and threshold $t \geq 0$, $\mathcal{S}(v,t)$ denotes coordinatewise soft-thresholding, $\mathcal{S}(v,t)_k = \operatorname{sign}(v_k)\max(|v_k| - t, 0)$, which is the proximal operator of $t\|\cdot\|_1$ (Parikh and Boyd, 2014). We write $\bar{\lambda} = \sum_j w_j \lambda_j$ and $\operatorname{supp}(v) = \{k : v_k \neq 0\}$.

**Lemma 1** (The weighted objective is a pooled Lasso)**.** *With $w_j = m_j/m$,*

$$F(\beta) = \frac{1}{2m}\|y - X\beta\|_2^2 + \bar{\lambda}\|\beta\|_1.$$

**Algorithm 1. Synchronous federated CD-Lasso**

1. Input local data $(X_j, y_j)$, fixed penalties $\lambda_j$, weights $w_j$, local epochs $E \geq 1$, and tolerance $\varepsilon_{\text{tol}} > 0$. Initialise $\beta_0 = 0$.
2. At round $t$, the server broadcasts $\beta_t$.
3. Each site in parallel computes $\beta_{j,t} = T_j^E(\beta_t)$ and uploads this coefficient vector.
4. The server forms $\beta_{t+1} = \sum_j w_j \beta_{j,t}$.
5. Stop if $\|\beta_{t+1} - \beta_t\|_2 < \varepsilon_{\text{tol}}$; otherwise increase $t$ and repeat from step 2. A finite cap is specified for each experiment.

*Proof.* Substitute $w_j = m_j/m$ into (2) and sum the quadratic terms and penalties separately. □

Lemma 1 fixes the objective and penalty when comparing aggregation with an exact consensus solver. Throughout, $\hat{\beta}_F$ denotes a minimiser of (2); uniqueness holds under the orthogonal assumptions below. A positive value of $F(\beta) - F(\hat{\beta}_F)$ measures optimisation error for this objective. It does not by itself imply worse support recovery or test prediction. The averaging protocol need not minimise $F$, as Proposition 3 shows.

### 2.2 The protocol

Algorithm 1 states the procedure studied here. The local operator is one or more full cyclic coordinate-descent epochs on (1), warm-started at the broadcast vector. For coordinate $k$ at site $j$, with $r = y_j - X_j\beta + x_{jk}\beta_k$,

$$\beta_k \;\leftarrow\; \frac{\mathcal{S}\Big(x_{jk}^\top r,\; \lambda_j m_j\Big)}{\|x_{jk}\|_2^2}, \tag{3}$$

the factor $m_j$ appearing because (1) is written on the normalised scale while (3) uses unnormalised residuals (Friedman et al., 2010). One epoch is one sweep over all $p$ coordinates. Let $T_j^E : \mathbb{R}^p \to \mathbb{R}^p$ denote the map given by $E$ such epochs at site $j$.

For a single fitted path, upload cost is $R \times K \times p \times 8$ bytes after $R$ rounds: coefficient vectors are transmitted in a dense double-precision representation, including their zero entries. Local work is $R \times E$ coordinate-descent epochs per site. These measures exclude download traffic and are not wall-clock timings. Tuning that requires additional federated training paths adds their communication and computation to the total; validation summaries are counted separately. We report both upload and local work because fewer rounds need not imply less computation.

## 3 Aggregation, sparsity and fixed points

We first analyse an exactly solvable orthogonal design, then characterise fixed points for general designs. The orthogonal results distinguish support enlargement caused by averaging from dependence on the number of local epochs. They do not assume that every average of sparse vectors is dense.

### 3.1 Orthogonal designs

Assume in this subsection that each site satisfies $X_j^\top X_j = m_j I_p$, which requires $p \le \min_j m_j$, and write $z_j = X_j^\top y_j / m_j$. Penalties are treated as fixed positive constants.

**Proposition 2** (Orthogonal designs)**.** *Under these assumptions:*

*(i) For every integer $E \ge 1$ and starting point $\beta \in \mathbb{R}^p$, $T_j^E(\beta) = \mathcal{S}(z_j, \lambda_j)$.*

*(ii) The unrestricted sequence in Algorithm 1 satisfies $\beta_t = \bar\beta := \sum_j w_j \mathcal{S}(z_j, \lambda_j)$ for all $t \ge 1$. With $\varepsilon_{\mathrm{tol}} > 0$, the stopping rule terminates within two rounds. Its output does not depend on $E$.*

*(iii) The pooled design satisfies $X^\top X = mI_p$, and its unique minimiser is $\hat\beta_F = \mathcal{S}(\bar z, \bar\lambda)$, where $\bar z = \sum_j w_j z_j$.*

*(iv) The supports satisfy $\operatorname{supp}(\hat\beta_F) \subseteq \bigcup_j \operatorname{supp}(\mathcal{S}(z_j, \lambda_j))$. For fixed penalties and weights, this union equals $\operatorname{supp}(\bar\beta)$ except on a Lebesgue-null subset of $(z_1, \dots, z_K) \in \mathbb{R}^{Kp}$ where non-zero contributions cancel exactly.*

*Proof.* For coordinate $k$, orthogonality gives

$$x_{jk}^\top (y_j - X_j\beta + x_{jk}\beta_k) = x_{jk}^\top y_j, \qquad \|x_{jk}\|_2^2 = m_j.$$

Equation (3) therefore sets the coordinate to $\mathcal{S}(z_{jk}, \lambda_j)$ independently of its starting value. One sweep sets every coordinate to its local minimiser and subsequent sweeps leave it unchanged. This proves (i). Consequently $\beta_1 = \beta_2 = \bar\beta$, proving (ii); the stopping rule may already hold in round one. For (iii), sum $X_j^\top X_j = m_j I_p$ over sites and apply Lemma 1 coordinatewise.

If $|z_{jk}| \le \lambda_j$ at every site, then $|\bar z_k| \le \sum_j w_j |z_{jk}| \le \bar\lambda$, so coordinate $k$ is absent from the pooled solution. This proves the inclusion in (iv). To prove its second statement, partition $\mathbb{R}^{Kp}$ by the local active sets and signs. On each of the finitely many resulting regions, $\bar\beta_k$ is affine in the $z_{jk}$. If at least one site is active at coordinate $k$, the coefficient of that site's $z_{jk}$ is $w_j > 0$. Thus $\bar\beta_k = 0$ lies on a proper affine hyperplane. The union of these hyperplanes over coordinates and regions, together with their boundaries, has Lebesgue measure zero. □

The support inclusion need not be strict. For example, with $K = 2$, equal weights, $\lambda_1 = \lambda_2 = 1$, $z_1 = (2, 0, 0)$ and $z_2 = (3, 0, 0)$, the average and pooled solution both equal $(1.5, 0, 0)$. Their supports remain equal and sparse on an open neighbourhood of these statistics. Averaging can enlarge a support when sites select different coordinates; it does not necessarily remove all exact zeros.

Part (ii) establishes exact independence from $E$ for orthogonal local designs. Nonorthogonal designs can require multiple coordinate-descent sweeps and hence can produce $E$-dependent iterates. This observation does not provide a quantitative ordering across correlated designs. In particular, independent predictors in the population do not yield orthogonal empirical Gram matrices when $p > m_j$.

**Corollary 1** (False-positive probability under a fixed penalty)**.** *In the setting of Proposition 2, suppose $K \ge 2$, $\lambda_j = \lambda > 0$, and, for a null coordinate $k$, $z_{jk} \sim N(0, \sigma^2/m_j)$ independently across sites, with $\sigma > 0$. Then*

$$\mathbb{P}(k \in \operatorname{supp}(\bar\beta)) = 1 - \prod_{j=1}^{K} \{1 - 2\bar\Phi(\lambda\sqrt{m_j}/\sigma)\} > 2\bar\Phi(\lambda\sqrt{m}/\sigma) = \mathbb{P}(k \in \operatorname{supp}(\hat\beta_F)),$$

*where $\bar{\Phi}$ is the standard normal survival function. For fixed $m$, $\lambda$ and $\sigma$, the left-hand side strictly increases with $K$ along balanced partitions $m_j = m/K$ satisfying the design assumptions.*

*Proof.* The Gaussian statistics have a joint density, so cancellation in Proposition 2(iv) has probability zero. Independence of the events $\{|z_{jk}| > \lambda\}$ gives the first equality. Their union has probability at least $2\bar{\Phi}(\lambda\sqrt{m_1}/\sigma)$, which strictly exceeds $2\bar{\Phi}(\lambda\sqrt{m}/\sigma)$ because $m_1 < m$. Also $\bar{z}_k \sim N(0, \sigma^2/m)$, giving the pooled probability. For balanced partitions let $q_K = 2\bar{\Phi}(\lambda\sqrt{m/K}/\sigma)$. Both $q_K \in (0, 1)$ and $1 - (1 - q_K)^K$ strictly increase with $K$. □

The corollary concerns exact non-zero coefficients and a fixed common penalty. It does not establish the same monotonicity for independently tuned penalties, arbitrary repartitions, or coefficients declared active only above a positive numerical threshold.

## 3.2 General designs: fixed points

Write $D_j^E(b) = T_j^E(b) - b$ for the local displacement. The following statement characterises fixed points; it does not assert that the averaged iteration converges to one.

**Proposition 3** (Averaged stationarity)**.** *A point $b^\dagger$ is a fixed point of Algorithm 1 if and only if*

$$\sum_{j=1}^{K} w_j D_j^E(b^\dagger) = 0. \tag{4}$$

*All its local displacements vanish if and only if $b^\dagger \in \bigcap_j \arg\min \mathcal{L}_j$. Thus, if this intersection is empty, any fixed point has non-zero local displacements that cancel in the weighted average. In general,* (4) *does not imply* $0 \in \partial F(b^\dagger)$.

*Proof.* The update equals $b + \sum_j w_j D_j^E(b)$, proving the first equivalence. For a fixed site, each coordinate subproblem is strictly convex because its column has positive squared norm. An exact coordinate update therefore strictly decreases $\mathcal{L}_j$ whenever it changes the coordinate. If $T_j^E(b) = b$, the objective has the same value before and after the sweeps, so no update can have changed a coordinate. The coordinatewise optimality conditions then give $0 \in X_j^\top(X_j b - y_j)/m_j + \lambda_j \partial\|b\|_1$, which is sufficient for a global minimum by convexity. Conversely, a minimiser satisfies these conditions and every unique coordinate update leaves it fixed. This proves the second equivalence.

For the final nonimplication take $K = 2$, $p = m_1 = m_2 = 1$, $X_1 = X_2 = [1]$, $y_1 = 2$, $y_2 = 0$ and $\lambda_1 = \lambda_2 = 1$. For every $E \geq 1$ the local maps return 1 and 0, respectively, so $b^\dagger = 1/2$ is an averaged fixed point. The pooled minimiser is $\mathcal{S}(1, 1) = 0$, while $F'(1/2) = 1/2 \neq 0$. □

Distinct local minimisers do not by themselves exclude a pooled-optimal fixed point; the sparse example following Proposition 2 already shows this. The relevant diagnostic is the objective gap to a numerically converged solution of the same pooled problem, rather than the local displacements alone. Consensus ADMM provides that comparison without requiring row-level exchange (Boyd et al., 2011).

**Remark 1.** *Proposition 2 requires $p \leq \min_j m_j$ and does not apply directly to our $p = 600 > m_j$ simulations. The orthogonal experiment in Supplement S1, Table S3, directly illustrates Proposition 2 and Corollary 1. Proposition 3 applies to general designs but gives neither a convergence rate nor a bound on the objective gap as a function of $E$. The experiments assess support, optimisation accuracy and computational cost separately within their specified designs.*

# 4 Selection steps and comparison methods

Proposition 2 explains why averaging can retain more variables than the pooled Lasso. We evaluate two thresholding variants and distinguish their selection properties from minimisation of the objective $F$. These are retrofits to CD averaging, not new composite optimisation algorithms. Section 7 separately evaluates adapted FedDualAvg as an established alternative for the common-penalty objective.

## 4.1 FedAvg-ST: thresholding after convergence

Let $\beta_R$ be the output of Algorithm 1. For each candidate $\tau \in \mathcal{T}$, a site evaluates $\mathcal{S}(\beta_R, \tau)$ on its validation data and returns

$$A_j(\tau) = \sum_{i=1}^{m_j^{\text{val}}} r_{ji}(\tau)^2, \qquad B_j(\tau) = \sum_{i=1}^{m_j^{\text{val}}} r_{ji}(\tau)^4, \qquad m_j^{\text{val}}, \tag{5}$$

where $r_{ji}(\tau) = y_{ji} - x_{ji}^\top \mathcal{S}(\beta_R, \tau)$. The server obtains $\widehat{\text{MSE}}(\tau) = \sum_j w_j A_j(\tau)/m_j^{\text{val}}$. All experiments use validation shares satisfying $w_j = m_j^{\text{val}}/n_{\text{val}}$, where $n_{\text{val}} = \sum_j m_j^{\text{val}}$. The pooled observation-level standard-error estimate is consequently

$$\widehat{\text{SE}}(\tau)^2 = \frac{\sum_j B_j(\tau) - (\sum_j A_j(\tau))^2/n_{\text{val}}}{n_{\text{val}}(n_{\text{val}} - 1)}.$$

For heterogeneous fixed sites this is a pooled one-standard-error heuristic, not a site-stratified variance estimator or a calibrated confidence interval. The selected threshold is

$$\hat{\tau} = \max\{\tau \in \mathcal{T} : \widehat{\text{MSE}}(\tau) \le \widehat{\text{MSE}}(\tau_{\min}) + \widehat{\text{SE}}(\tau_{\min})\}, \qquad \hat{\beta}^{\text{ST}} = \mathcal{S}(\beta_R, \hat{\tau}), \tag{6}$$

where $\tau_{\min}$ minimises validation MSE. The largest eligible threshold favours a smaller active set. We also report FedAvg-ST-min, which uses $\tau_{\min}$. Neither rule uses true support or independent test data.

Post-fit selection adds one validation exchange. Counting the sample count for every candidate gives $3K|\mathcal{T}|$ uploaded double-precision scalars, or 2.81 KiB for $K = 3$ and $|\mathcal{T}| = 40$. Every site must also receive the final coefficient vector and grid; downloads are excluded from the upload convention.

## 4.2 FedAvg-P: thresholding in each round

The per-round variant uses $\beta_{t+1} = \mathcal{S}(\sum_j w_j \beta_{j,t}, \tau)$. For each $\tau$ in $\bar{\lambda}\{1, 1/2, 1/4, 1/8, 1/16\}$, we fit a separate trajectory and select the largest threshold within one SE of minimum validation error. This is a heuristic sparsification of the CD iteration; it does not inherit convergence guarantees for $F$ from FedDualAvg or other composite methods (Yuan et al., 2021; Bao et al., 2022). If candidate $q$ takes $R_q$ rounds, full vector upload is $8Kp\sum_{q=1}^5 R_q$ bytes and CD work is $E\sum_{q=1}^5 R_q$ sweeps per site. An additional 0.35 KiB covers validation scalars. The selected fit's cost is retained separately. Unlike post-fit ST, selecting P requires multiple complete fits.

## 4.3 Comparators

**Pooled and local Lasso.** The pooled reference uses all training records and tunes its penalty on the pooled validation sample. The local reference uses the largest site alone. Both are fitted under minimum-MSE and one-SE penalty selection. Pooled data are used for reference computation in this single-machine simulation; the federated iteration does not require them.

**One-shot averaging.** Each site fits its Lasso to convergence, followed by one weighted average. This is the large-local-work limiting comparator, with one vector upload per site. Both local penalty rules are included. The uncorrected average here differs from debiased distributed estimators such as those of Lee et al. (2017) and Battey et al. (2018).

**Consensus ADMM.** Consensus ADMM (Boyd et al., 2011) alternates local ridge updates, a server soft-threshold, and local dual updates. Each site uploads one primal-plus-dual vector per iteration. By Lemma 1, its target equals the pooled Lasso at $\bar{\lambda}$. We check numerical residuals rather than equating a finite iteration with an exact solution. This comparator holds $F$ fixed while changing the solver.

ADMM-CV and ADMM-CV-1SE instead tune the common penalty over a 15-point validation grid. We compute these paths using the equivalent pooled objective. Their statistical performance is reported, but no unmeasured federated tuning cost is assigned to these centrally computed paths.

### 4.4 Tuning comparisons

The -1SE suffix denotes the largest eligible penalty on the same grid used for minimum-MSE tuning. In particular, FedAvg-1SE changes each local penalty before averaging, and One-shot-1SE changes each converged local fit. These comparisons isolate the local penalty rule within an aggregation method. ST and P instead use minimum-MSE local penalties followed by one-SE server-threshold selection. They are distinct two-stage procedures; sharing the name "one-SE" does not make their search spaces identical. ADMM at inherited $\bar{\lambda}$ is a fixed-objective control, not a one-SE-tuned method. Consequently, ST versus FedAvg-1SE measures the performance of two different tuning procedures. It does not isolate a server-threshold effect at identical local penalties and identical tuning opportunity. Likewise, the fixed-penalty ADMM comparison evaluates a solver at an inherited penalty; its support performance does not establish a tuning advantage.

### 4.5 Metrics and cost conventions

A coordinate is active when $|\hat{\beta}_k| > \epsilon$, with $\epsilon = 10^{-4}$. Simulation precision, recall, $F_1$, and Jaccard index compare this set with known true support. Precision and $F_1$ are set to zero for an empty selected model, so all simulation means use all 200 replicates. Estimation error is $\|\hat{\beta} - \beta^\star\|_2$; test MSE uses independent observations. When true coefficients are unavailable, overlap with a pooled fit measures agreement, not correctness.

Fit-round totals include every P candidate; validation adds one further communication round. We distinguish uploaded vectors from validation scalars. Downloads, protocol headers, and local penalty-search work are excluded; these are not end-to-end runtime measurements. CD epochs quantify work within the CD family and do not measure ADMM's factorisation and linear solves. No formal privacy guarantee is claimed.

## 5 Simulation study

### 5.1 Data-generating process

Each replicate draws $\beta^\star \in \mathbb{R}^{600}$ with 30 nonzero entries at uniformly sampled positions. Their signs are independent and equally likely, and their magnitudes are uniform on $[0.4, 1.2]$. The three sites have training sizes $(400, 240, 160)$ and validation sizes $(100, 60, 40)$. Thus $m = 800$ and

$p = 600 > m_j$ at every site. Local Lasso fits remain well-defined, although a full-rank unpenalised local fit is unavailable. Responses satisfy $y_j = X_j\beta^\star + \epsilon_j$.

**Independent.** Design entries are independent standard normal.

**Correlated.** Rows have AR(1) Gaussian covariance with correlation 0.5.

**Heterogeneous.** Site correlations are $(0, 0.4, 0.7)$, design scale factors $(1, 1.4, 0.7)$, and noise scale multipliers $(1, 1.5, 0.8)$. The coefficient vector remains common across sites.

Let $v_j$ be the realised variance of the training signal $X_j\beta^\star$, calculated with denominator $m_j$. Gaussian noise has standard deviation $\sqrt{v_j}$ in the homogeneous designs and $c_j\sqrt{v_j}$ in the heterogeneous design, with $c = (1, 1.5, 0.8)$. The baseline signal-to-noise ratio is one before these noise multipliers. Training and validation responses share the site noise scale. Aggregation weights are $(0.5, 0.3, 0.2)$.

The independent test sample has 2000 observations. Its design has unit marginal scale and AR(1) correlation 0, 0.5, or 0.4 for the independent, correlated, or heterogeneous design, respectively. Its noise standard deviation is the unweighted mean of the three realised site standard deviations. Thus the heterogeneous test population is a specified common target, not a sample-size-weighted mixture of training sites.

Population independence does not make a finite high-dimensional design orthogonal. Proposition 2 supplies an exactly solvable reference case, not an approximation theorem for these simulations. The designs do not isolate every source of an epoch effect.

## 5.2 Protocol and tuning

Local grids contain 30 logarithmically spaced penalties from $\lambda_j^{\max} = \|X_j^\top y_j\|_\infty/m_j$ to $10^{-3}\lambda_j^{\max}$. The pooled reference uses the corresponding pooled grid. Penalties are fixed before fitting. Federated iterations start at zero and use $E \in \{1, 2, 3, 5, 8, 10, 15, 20\}$, stopping at an update norm below $10^{-5}$ or after 500 rounds. The ST grid is zero plus 39 logarithmically spaced thresholds on $[10^{-3}\bar{\lambda}, 5\bar{\lambda}]$; the P grid is $\bar{\lambda}\{1, 1/2, 1/4, 1/8, 1/16\}$.

Consensus ADMM uses augmented-Lagrangian parameter 1. Its primal residual is $(\sum_j \|x_j - z\|_2^2)^{1/2}$ and its server-update residual is $\|z^{\text{new}} - z^{\text{old}}\|_2$; both must be below $10^{-8}\sqrt{p}$, with a 500-iteration cap. The usual stacked dual residual is $\sqrt{K}$ times this server-update residual. The centrally computed ADMM-CV path contains 15 logarithmically spaced penalties on $[\bar{\lambda}/8, 4\bar{\lambda}]$.

## 5.3 Implementation and replication

The implementation uses cyclic coordinate descent in scikit-learn (Pedregosa et al., 2011). Warm-started local calls use zero convergence tolerance and a maximum of $E$ iterations; independent checks compare their output with explicit coordinate sweeps. Penalty paths use tolerance $10^{-4}$, final local fits $10^{-10}$, and the centrally computed consensus path $10^{-6}$. Convergence diagnostics and software versions accompany the results.

For each design, the 200 replicates use the same set of 200 seeds: 5000–5066 (67 seeds), 6000–6066 (67 seeds), and 7000–7065 (66 seeds). Together, these three blocks form the complete seed set for every design. All methods share the same data within a replicate. Monte Carlo standard errors are sample standard deviations divided by $\sqrt{200}$; schedule contrasts are calculated within replicate before averaging. Seeds are reused across designs, so design summaries are not independent experimental samples. The scripts regenerate results, tables, figures, and numerical macros for the main simulation study.

# 6 Simulation results

Table 1 summarises the three designs; the full epoch grid and cost accounting appear in Supplementary Table S1. All means include 200 replicates per design. Figure 1 separates changes in selection quality from changes in the number of selected variables.

**Table 1:** Simulation results, 200 replicates per design ($p = 600$, $s = 30$). $F_1$ is relative to the true support; parentheses give Monte Carlo standard errors for $F_1$ and test MSE. Precision is zero for empty selected models. The -1SE suffix denotes one-standard-error penalty selection; ST and P use one-SE threshold selection after local minimum-MSE penalty selection. ADMM uses the inherited fixed penalty $\bar{\lambda}$. Cost accounting is reported separately in Supplementary Table S1.

| Design | Method | Active | Precision | Recall | $F_1$ (MCSE) | Test MSE (MCSE) |
|---|---|---|---|---|---|---|
| Independent | Pooled | 120.6 | 0.258 | 0.979 | 0.405 (0.006) | 24.36 (0.20) |
| | Pooled-1SE | 48.1 | 0.601 | 0.915 | 0.715 (0.006) | 25.96 (0.22) |
| | Local-only | 107.3 | 0.264 | 0.862 | 0.395 (0.006) | 28.79 (0.24) |
| | Local-only-1SE | 39.0 | 0.579 | 0.664 | 0.582 (0.008) | 31.79 (0.32) |
| | One-shot | 194.0 | 0.155 | 0.940 | 0.264 (0.004) | 29.15 (0.24) |
| | One-shot-1SE | 53.5 | 0.470 | 0.739 | 0.547 (0.008) | 34.23 (0.30) |
| | ADMM | 41.6 | 0.709 | 0.856 | 0.743 (0.008) | 27.63 (0.31) |
| | ADMM-CV | 119.9 | 0.260 | 0.978 | 0.407 (0.006) | 24.35 (0.20) |
| | ADMM-CV-1SE | 48.7 | 0.602 | 0.918 | 0.715 (0.006) | 25.98 (0.22) |
| | FedAvg ($E = 1$) | 247.2 | 0.129 | 0.957 | 0.225 (0.004) | 28.85 (0.24) |
| | FedAvg ($E = 20$) | 197.7 | 0.154 | 0.941 | 0.262 (0.004) | 29.14 (0.24) |
| | FedAvg-1SE ($E = 1$) | 61.0 | 0.426 | 0.749 | 0.516 (0.008) | 34.19 (0.30) |
| | FedAvg-1SE ($E = 20$) | 53.5 | 0.470 | 0.739 | 0.547 (0.008) | 34.23 (0.30) |
| | FedAvg-ST ($E = 1$) | 43.2 | 0.612 | 0.789 | 0.667 (0.007) | 31.23 (0.28) |
| | FedAvg-ST ($E = 20$) | 44.9 | 0.588 | 0.785 | 0.649 (0.007) | 31.57 (0.27) |
| | FedAvg-P ($E = 1$) | 62.1 | 0.458 | 0.807 | 0.557 (0.008) | 31.06 (0.28) |
| | FedAvg-P ($E = 20$) | 50.5 | 0.550 | 0.799 | 0.623 (0.008) | 31.34 (0.28) |
| Correlated | Pooled | 120.8 | 0.254 | 0.966 | 0.398 (0.005) | 24.35 (0.20) |
| | Pooled-1SE | 51.5 | 0.553 | 0.896 | 0.673 (0.006) | 25.99 (0.23) |
| | Local-only | 102.3 | 0.267 | 0.838 | 0.397 (0.005) | 28.54 (0.24) |
| | Local-only-1SE | 41.4 | 0.533 | 0.652 | 0.552 (0.006) | 31.34 (0.31) |
| | One-shot | 190.2 | 0.154 | 0.925 | 0.261 (0.003) | 28.67 (0.24) |
| | One-shot-1SE | 58.5 | 0.426 | 0.732 | 0.514 (0.006) | 33.72 (0.30) |
| | ADMM | 45.3 | 0.633 | 0.845 | 0.697 (0.007) | 27.26 (0.29) |
| | ADMM-CV | 118.7 | 0.261 | 0.967 | 0.406 (0.006) | 24.37 (0.20) |
| | ADMM-CV-1SE | 52.2 | 0.545 | 0.897 | 0.668 (0.006) | 25.94 (0.22) |
| | FedAvg ($E = 1$) | 250.2 | 0.123 | 0.942 | 0.215 (0.004) | 28.34 (0.24) |
| | FedAvg ($E = 20$) | 193.0 | 0.153 | 0.926 | 0.260 (0.004) | 28.66 (0.24) |
| | FedAvg-1SE ($E = 1$) | 70.7 | 0.368 | 0.743 | 0.467 (0.007) | 33.68 (0.30) |
| | FedAvg-1SE ($E = 20$) | 58.5 | 0.426 | 0.732 | 0.513 (0.006) | 33.72 (0.30) |
| | FedAvg-ST ($E = 1$) | 53.6 | 0.503 | 0.788 | 0.590 (0.007) | 30.83 (0.28) |
| | FedAvg-ST ($E = 20$) | 55.3 | 0.479 | 0.783 | 0.572 (0.007) | 31.14 (0.28) |
| | FedAvg-P ($E = 1$) | 70.2 | 0.395 | 0.793 | 0.503 (0.007) | 30.68 (0.28) |
| | FedAvg-P ($E = 20$) | 59.7 | 0.457 | 0.793 | 0.556 (0.007) | 31.00 (0.28) |
| Heterogeneous | Pooled | 117.5 | 0.253 | 0.891 | 0.386 (0.006) | 38.10 (0.32) |
| | Pooled-1SE | 32.3 | 0.657 | 0.622 | 0.596 (0.009) | 42.44 (0.43) |
| | Local-only | 107.3 | 0.264 | 0.862 | 0.395 (0.006) | 38.92 (0.32) |
| | Local-only-1SE | 39.0 | 0.579 | 0.664 | 0.582 (0.008) | 41.94 (0.39) |
| | One-shot | 181.7 | 0.162 | 0.922 | 0.273 (0.004) | 40.00 (0.33) |
| | One-shot-1SE | 55.1 | 0.458 | 0.721 | 0.523 (0.007) | 44.74 (0.38) |
| | ADMM | 37.3 | 0.650 | 0.597 | 0.529 (0.011) | 42.69 (0.46) |
| | ADMM-CV | 119.1 | 0.248 | 0.892 | 0.381 (0.006) | 38.11 (0.32) |
| | ADMM-CV-1SE | 33.1 | 0.648 | 0.629 | 0.596 (0.009) | 42.33 (0.44) |
| | FedAvg ($E = 1$) | 249.5 | 0.125 | 0.941 | 0.218 (0.004) | 39.86 (0.33) |
| | FedAvg ($E = 20$) | 185.3 | 0.161 | 0.923 | 0.271 (0.004) | 39.99 (0.33) |
| | FedAvg-1SE ($E = 1$) | 69.1 | 0.396 | 0.739 | 0.475 (0.008) | 44.73 (0.38) |

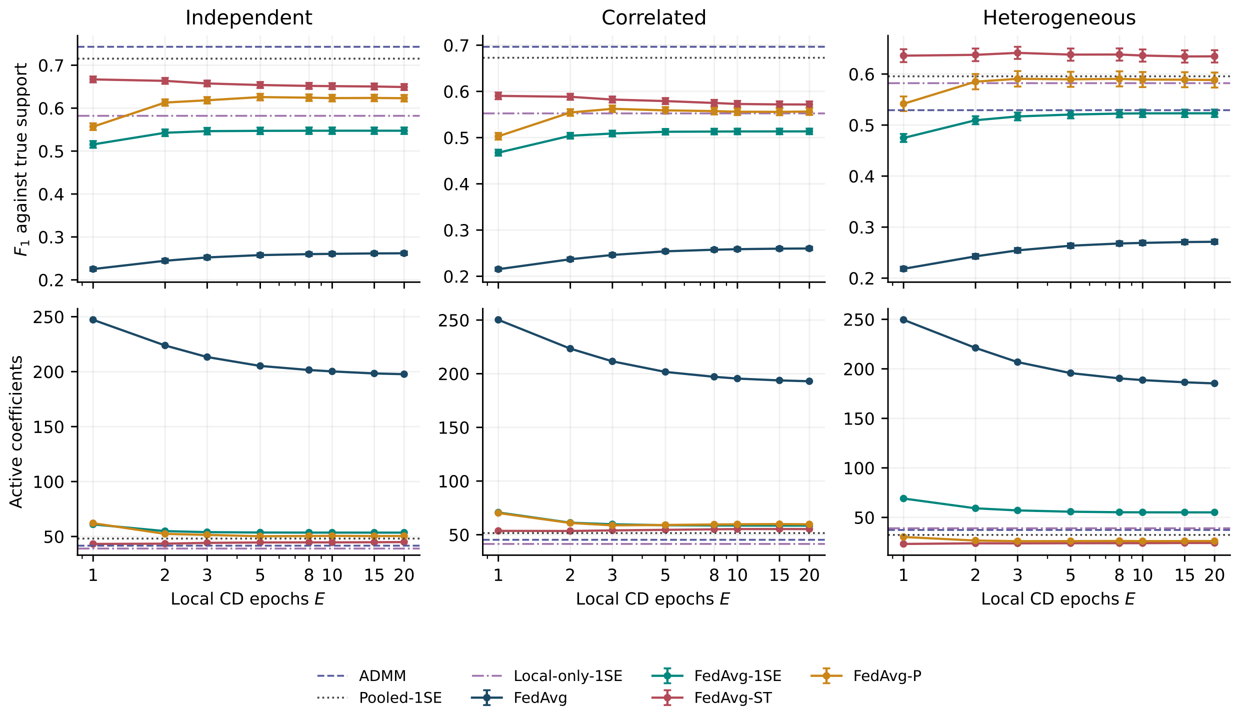


**Figure 1:** True-support $F_1$ (top) and active-set size (bottom) versus local CD epochs. Error bars are Monte Carlo standard errors. FedAvg uses local minimum-MSE penalties, FedAvg-1SE uses local one-SE penalties, and ST/P select a server threshold by one SE after minimum-MSE local tuning. ADMM uses inherited $\bar{\lambda}$; the other horizontal references use one-SE penalty selection. The true support contains 30 of 600 variables.

| Design | Method | Active | Precision | Recall | $F_1$ (MCSE) | Test MSE (MCSE) |
|---|---|---|---|---|---|---|
| | FedAvg-1SE ($E = 20$) | 55.1 | 0.458 | 0.721 | 0.523 (0.007) | 44.74 (0.38) |
| | FedAvg-ST ($E = 1$) | 23.2 | 0.787 | 0.579 | 0.636 (0.013) | 44.99 (0.41) |
| | FedAvg-ST ($E = 20$) | 24.2 | 0.787 | 0.585 | 0.635 (0.012) | 45.14 (0.41) |
| | FedAvg-P ($E = 1$) | 30.2 | 0.616 | 0.562 | 0.542 (0.014) | 45.04 (0.44) |
| | FedAvg-P ($E = 20$) | 26.1 | 0.707 | 0.570 | 0.588 (0.015) | 45.13 (0.43) |

## 6.1 Averaging and local penalty selection

With minimum-MSE local penalties, FedAvg selects 247, 250, and 250 variables at $E = 1$ in the independent, correlated, and heterogeneous designs. Its recalls are 0.957, 0.942, and 0.941, but precisions are only 0.129, 0.123, and 0.125. The average retains substantially more variables than the true model, consistent with support enlargement through aggregation.

Increasing $E$ from 1 to 20 reduces active-set size by 20.0%, 22.9%, and 25.7%, and increases $F_1$ from 0.225 to 0.262, 0.215 to 0.260, and 0.218 to 0.271. The large-$E$ results approach one-shot averaging, whose $F_1$ values are 0.264, 0.261, and 0.273. The Gram matrices are nonorthogonal, the site fits differ, and the aggregation map changes with $E$; these effects occur together.

Changing the local penalty rule improves averaging without adding a server threshold. FedAvg-1SE gives $F_1 = 0.516$, 0.467, and 0.475 at $E = 1$, and 0.547, 0.513, and 0.523 at $E = 20$. One-shot-1SE gives 0.547, 0.514, and 0.523. These controls show why a comparison between prediction-tuned averaging and sparsity-favouring thresholding cannot attribute the entire difference to aggregation alone.

**Table 2:** Paired change in true-support $F_1$, $E = 20$ minus $E = 1$. Each difference is computed within replicate before averaging; MCSE is the standard deviation of the paired differences divided by $\sqrt{200}$.

| Design | Method | Mean change | Paired MCSE |
|---|---|---|---|
| Independent | FedAvg | +0.0367 | 0.0012 |
| | FedAvg-1SE | +0.0319 | 0.0022 |
| | FedAvg-ST | -0.0178 | 0.0044 |
| | FedAvg-P | +0.0663 | 0.0066 |
| Correlated | FedAvg | +0.0449 | 0.0011 |
| | FedAvg-1SE | +0.0460 | 0.0024 |
| | FedAvg-ST | -0.0187 | 0.0051 |
| | FedAvg-P | +0.0534 | 0.0061 |
| Heterogeneous | FedAvg | +0.0530 | 0.0017 |
| | FedAvg-1SE | +0.0484 | 0.0029 |
| | FedAvg-ST | -0.0013 | 0.0055 |
| | FedAvg-P | +0.0466 | 0.0084 |

### 6.2 Thresholding, prediction and epoch effects

FedAvg-ST attains $F_1 = 0.667$ (MCSE 0.007), 0.590 (0.007), and 0.636 (0.013) at $E = 1$, with active-set sizes 43, 54, and 23. FedAvg-P attains 0.557, 0.503, and 0.542 at $E = 1$, rising to 0.623, 0.556, and 0.588 at $E = 20$. The two thresholding procedures therefore have different epoch responses.

Table 2 quantifies these effects using within-replicate contrasts. ST has a modest decline in the homogeneous designs that is resolved relative to its paired Monte Carlo uncertainty, while the heterogeneous-design contrast is much smaller. P has positive contrasts in all three designs. Thresholded selection therefore remains sensitive to $E$. Mean $F_1$ contrasts describe recovery quality, while within-replicate agreement requires comparing the selected variable identities.

Selection gains can come with increased estimation and prediction error. For example, in the independent design at $E = 1$, moving from FedAvg to ST changes coefficient error from 2.85 to 3.24 and test MSE from 28.8 to 31.2 (Supplementary Table S1). The preferred estimator therefore depends on whether variable selection, estimation, or prediction is the primary objective. These metrics should be reported together.

### 6.3 Optimisation accuracy and statistical performance

Pooled minimum-MSE tuning gives $F_1 = 0.405$, 0.398, and 0.386, compared with 0.715, 0.673, and 0.596 under one-SE tuning. The corresponding ADMM-CV values are 0.407, 0.406, and 0.381, versus 0.715, 0.668, and 0.596 with one SE. The tuning rule can change support recovery more than the epoch contrasts considered here.

ADMM at the inherited penalty gives $F_1 = 0.743$, 0.697, and 0.529. Its favourable selection results in the homogeneous designs depend partly on the inherited penalty. In the heterogeneous design, retuning the common penalty by one SE improves $F_1$ from 0.529 to 0.596. This does not establish that a common penalty is intrinsically inadequate for heterogeneous sites. In particular, ST's threshold is also common to all coordinates and sites, and the present experiments do not isolate a site-adaptation mechanism.

A converged pooled Lasso minimises its specified penalised training objective. It does not necessarily maximise $F_1$ or minimise test error. Consequently a thresholded estimator can outperform it on a statistical metric while having a larger value of $F$.

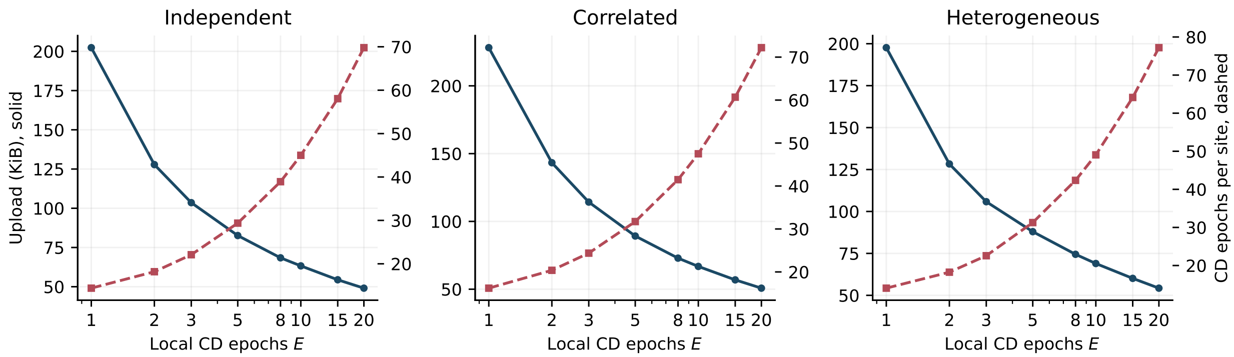


**Figure 2:** FedAvg upload (solid, left axis) and CD sweeps per site (dashed, right axis), averaged over 200 replicates. These are training-path costs, excluding initial local penalty search and downloads.

### 6.4 Communication and local work

For minimum-MSE FedAvg in the independent design, increasing $E$ from 1 to 20 reduces rounds from 14.4 to 3.5, but increases CD work per site from 14.4 to 69.8. The correlated and heterogeneous designs also show this trade-off (16.2 to 72.2, and 14.1 to 77.2 CD sweeps, respectively). A communication-efficient schedule need not be computation-efficient.

Post-fit ST adds only 2.8 KiB of validation scalars to its underlying fit. P instead requires all five candidate trajectories. In the independent design at $E = 1$, its selected fit averages 12.7 rounds, whereas all candidates total 60.7 rounds and 854.3 KiB including validation summaries. At $E = 20$ these totals are 15.8 rounds and 223.0 KiB. Reporting only the selected P fit would materially understate the cost of obtaining that estimator. ADMM uses 58, 92, and 66 rounds at its stricter residual tolerance; differing stopping criteria must be considered when interpreting these costs.

## 7 Comparison with adapted FedDualAvg at fixed selected-path uploads

We additionally compare coordinate-descent averaging with a deterministic specialisation of FedDualAvg (Yuan et al., 2021). This experiment uses the first 50 recorded seeds in each simulation design, 5000–5049. The penalty $\bar{\lambda}$ is selected as in Section 5, but every site uses this *common* penalty in both methods. Thus the objective and its local penalty decomposition are fixed across methods. The coordinate-descent arm is labelled CD-common to distinguish it from the locally tuned FedAvg fits in the main study. A converged pooled Lasso at $\bar{\lambda}$ supplies an optimisation reference, without being assigned a federated communication cost.

The adapted FedDualAvg follows Algorithm 3 of Yuan et al. (2021) with full participation, sample-size weights, full local empirical gradients, $h(b) = \|b\|_2^2/2$, $\psi(b) = \bar{\lambda}\|b\|_1$ and server learning rate one. Write $f_j(b) = \|y_j - X_j b\|_2^2/(2m_j)$ for the smooth local loss. At zero-based round $r$ and local step $k$, it forms $b_{j,r,k} = \mathcal{S}(z_{j,r,k}, \eta(rE + k)\bar{\lambda})$ and updates $z_{j,r,k+1} = z_{j,r,k} - \eta \nabla f_j(b_{j,r,k})$. The server averages the final dual states with weights $w_j$ and applies $\mathcal{S}(\,\cdot\,, \eta(r + 1)E\bar{\lambda})$ to retrieve its primal iterate. All dual states start at zero. We report the final server iterate, not the within-round averaged iterate used in some of that paper's theoretical bounds. A local step is a full-gradient evaluation from locally cached Gram and score matrices; it is not a coordinate-descent sweep. These matrices are not transmitted. We therefore match uploads, not local arithmetic or wall-clock time.

For each $E \in \{1, 5, 20\}$, both methods run without early stopping and are evaluated at $R \in \{5, 10, 20, 50, 100\}$ rounds. For adapted FedDualAvg, the client step is selected separately at each $(E, R)$ by minimum weighted validation MSE from $\eta \in \{0.1, 0.3, 1\}/L_{\max}$, where $L_{\max} = \max_j \lambda_{\max}(X_j^\top X_j/m_j)$ is calculated from one scalar supplied by each site. Ties favour the smaller step. Test responses and true support are not used for tuning. This finite grid and fixed server step define the implemented comparator; the experiment does not optimise over all possible FedDualAvg configurations.

One selected-fit round uploads $8Kp = 14{,}400$ bytes, so both displayed fits have the same selected-path vector upload at a given $R$. However, selecting the FedDualAvg step requires three fits. For a stand-alone budget $R$, its total upload is $3R(8Kp)+8K(2+3)$ bytes: three vector paths, the local penalties and Lipschitz constants, and three validation losses per site. The CD-common total is $R(8Kp) + 8K$ bytes. At $R = 100$ these are 4218.87 and 1406.27 KiB, respectively, versus 1406.25 KiB for either selected vector path. Local penalty-grid fits require local computation but no additional uploads. Downloads, transport overhead and offline diagnostic evaluation are excluded. The released cost ledger also accounts for shared setup and reused path prefixes across the whole grid; stand-alone budget costs must not be added together to represent that shared run.

We report true-support $F_1$, test MSE, objective difference from the pooled reference, and the infinity norm of the unit-step proximal-gradient mapping $b - \mathcal{S}(b - \sum_j w_j \nabla f_j(b), \bar{\lambda})$. A zero mapping is equivalent to the Lasso optimality conditions. These diagnostics distinguish optimisation agreement from support recovery. Table 3 reports the 100-round comparison and Figure 3 shows the budget trajectories. Full candidate results and paired Monte Carlo differences are included with the code.

At 100 rounds, the ranges of mean true-support $F_1$ across the three local-step settings are 0.732–0.749, 0.713–0.734, 0.518–0.549 for adapted FedDualAvg in the independent, correlated and heterogeneous designs, respectively; the corresponding CD-common ranges are 0.256–0.298, 0.235–0.282, 0.234–0.285. These comparisons condition on the selected step-size path; the additional tuning cost stated above remains part of the complete procedure.

The optimisation and selection comparisons point in the same direction: adapted FedDualAvg has smaller mean objective gaps and higher mean $F_1$ in every displayed design and local-step setting. For the independent design at $E = 1$, its mean gap is $2.7 \times 10^{-5}$, compared with 1.695 for CD-common. This is evidence for using the existing composite method when accurate minimisation of the common-penalty objective is required. The common-cap sensitivity analysis below additionally charges the tuning paths.

ST and P in Section 4 are workflow retrofits: ST adds a small validation exchange, whereas P trains five candidates. This comparison uses a common penalty and tunes a dual-averaging step, whereas the main study uses local penalties plus server-threshold tuning. A direct ranking of ST, P and FedDualAvg would require a separate experiment matching these choices and total costs.

**Sensitivity to complete tuning cost.** Supplement S1, Table S2, applies a common total-upload cap of 1,440,120 bytes (1406.37 KiB) to the archived runs, counting all candidate paths and the stand-alone setup and validation scalars above. For each method and $E$, the largest recorded affordable checkpoint is selected using cost alone: $R = 100$ for CD-common and $R = 20$ for adapted FedDualAvg. Actual uploads are 1406.27 and 843.87 KiB, respectively. The coarse checkpoint grid leaves unused budget, so this is a common-cap comparison rather than a comparison at equal spending. It was specified as a sensitivity analysis after the original runs; no new trajectories or interpolated outcomes are used. Adapted FedDualAvg retains smaller mean objective gaps and higher mean $F_1$ in all nine design–$E$ cells; paired mean $F_1$ differences range from 0.243 to 0.467.

**Table 3:** Fixed common-penalty comparison after 100 selected-fit upload rounds. Means (Monte Carlo standard errors) over 50 replicates per design. CD is averaging of coordinate-descent iterates with common penalty; DA is adapted FedDualAvg. DA's three-rate search costs 300 vector-upload rounds. $G_\infty$ is the unit-step proximal-gradient mapping residual.

| Design | Method | $E$ | Objective gap | $G_\infty$ | $F_1$ | Test MSE |
|---|---|---|---|---|---|---|
| Independent | CD | 1 | 1.695 (0.051) | 0.213 (0.006) | 0.256 (0.013) | 29.666 (0.564) |
| | CD | 5 | 1.614 (0.054) | 0.225 (0.006) | 0.295 (0.014) | 30.314 (0.558) |
| | CD | 20 | 1.642 (0.057) | 0.230 (0.006) | 0.298 (0.014) | 30.380 (0.559) |
| | DA | 1 | $2.7 \times 10^{-5}$ ($2.1 \times 10^{-5}$) | 0.002 (0.001) | 0.732 (0.017) | 28.021 (0.650) |
| | DA | 5 | $7.9 \times 10^{-4}$ ($2.1 \times 10^{-4}$) | 0.013 (0.001) | 0.741 (0.016) | 28.064 (0.650) |
| | DA | 20 | 0.003 (0.001) | 0.022 (0.002) | 0.749 (0.016) | 28.101 (0.653) |
| Correlated | CD | 1 | 1.665 (0.037) | 0.206 (0.007) | 0.235 (0.007) | 28.990 (0.518) |
| | CD | 5 | 1.530 (0.034) | 0.226 (0.008) | 0.278 (0.008) | 29.549 (0.511) |
| | CD | 20 | 1.548 (0.035) | 0.229 (0.008) | 0.282 (0.007) | 29.588 (0.511) |
| | DA | 1 | $8.7 \times 10^{-5}$ ($2.0 \times 10^{-5}$) | 0.004 (0.000) | 0.713 (0.012) | 27.414 (0.527) |
| | DA | 5 | 0.001 ($3.7 \times 10^{-4}$) | 0.014 (0.002) | 0.728 (0.011) | 27.454 (0.529) |
| | DA | 20 | 0.003 (0.001) | 0.021 (0.003) | 0.734 (0.011) | 27.478 (0.529) |
| Heterogeneous | CD | 1 | 3.122 (0.086) | 0.395 (0.016) | 0.234 (0.008) | 44.771 (0.822) |
| | CD | 5 | 2.448 (0.066) | 0.408 (0.016) | 0.277 (0.007) | 44.913 (0.804) |
| | CD | 20 | 2.475 (0.067) | 0.413 (0.017) | 0.285 (0.007) | 44.997 (0.798) |
| | DA | 1 | 0.002 (0.001) | 0.016 (0.006) | 0.518 (0.023) | 42.911 (0.937) |
| | DA | 5 | 0.015 (0.004) | 0.058 (0.006) | 0.539 (0.025) | 42.949 (0.938) |
| | DA | 20 | 0.039 (0.013) | 0.099 (0.011) | 0.549 (0.028) | 43.024 (0.943) |

Thus its advantage over CD-common is not confined to giving its selected path the same uploads while omitting the other tuning paths. These results still do not rank ST or P, or match local computation. Paired differences and their Monte Carlo standard errors are provided with the code.

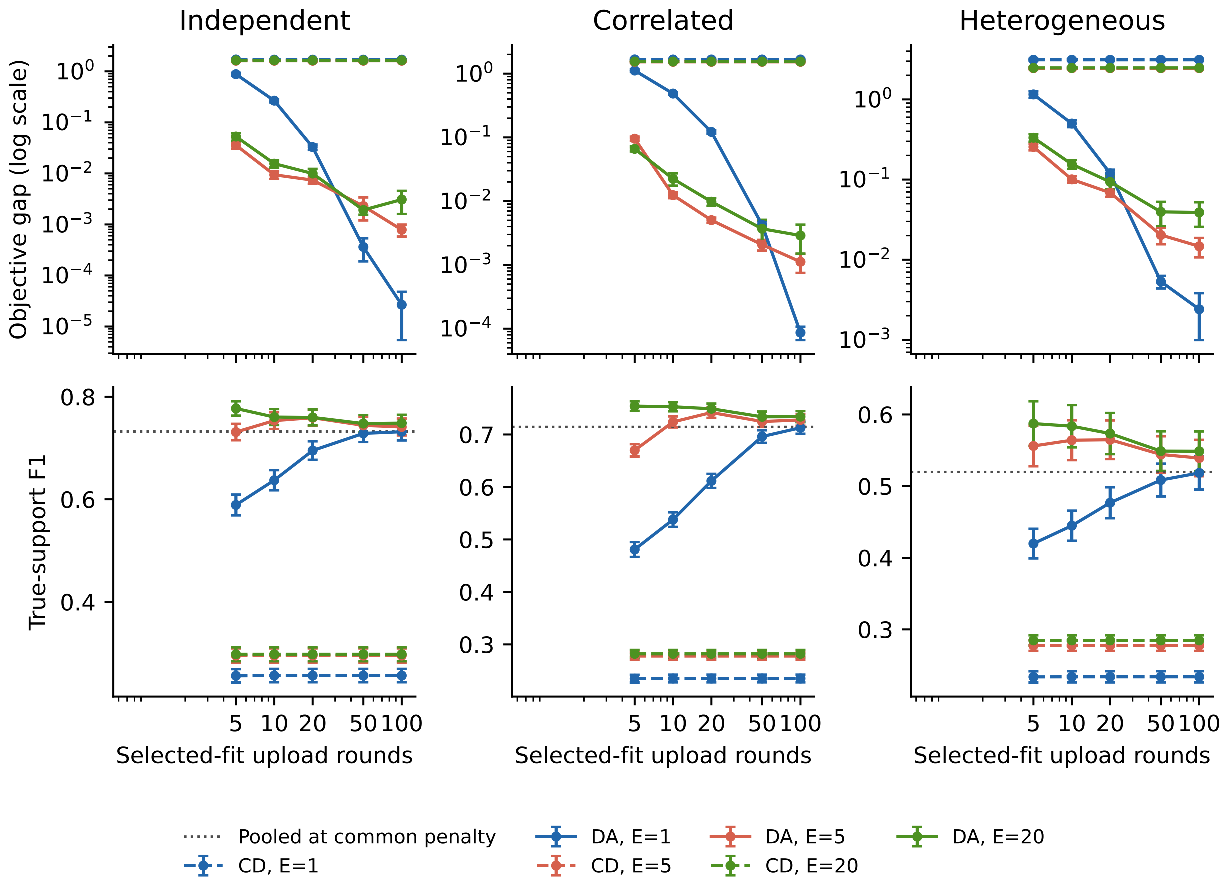


**Figure 3:** Common-penalty comparison at fixed selected-fit upload budgets. CD denotes CD-COMMON; DA denotes adapted FEDDUALAVG. Error bars are Monte Carlo standard errors over 50 paired replicates per design. DA's validation search requires three vector paths; the horizontal axis excludes those additional tuning paths. Dotted lines show the same-penalty pooled reference for $F_1$. Local steps differ between methods.

## 8 A fixed synthetic three-site benchmark

The supplied node files are synthetic. We retain this fixed benchmark with 800 training and 200 validation observations, 600 predictors, and site sizes $(400, 240, 160)$ and $(100, 60, 40)$, respectively. Its original generation procedure, seed and coefficient vector were not supplied, so the generating support cannot be evaluated. The matrices are used without further preprocessing. Their unknown generation details preclude claims about the site distributions. The original files and their SHA-256 checksums are retained with the code, so this analysis can be repeated from the fixed inputs even though new datasets from the original generator cannot be reproduced. This is a descriptive computational example, not an observational application. Agreement refers to the pooled minimum-MSE Lasso support, not correctness.

Local penalties are $(0.2046, 0.3635, 0.1953)$, giving $\bar{\lambda} = 0.2504$. The pooled reference selects penalty 0.1778 and retains 172 predictors.

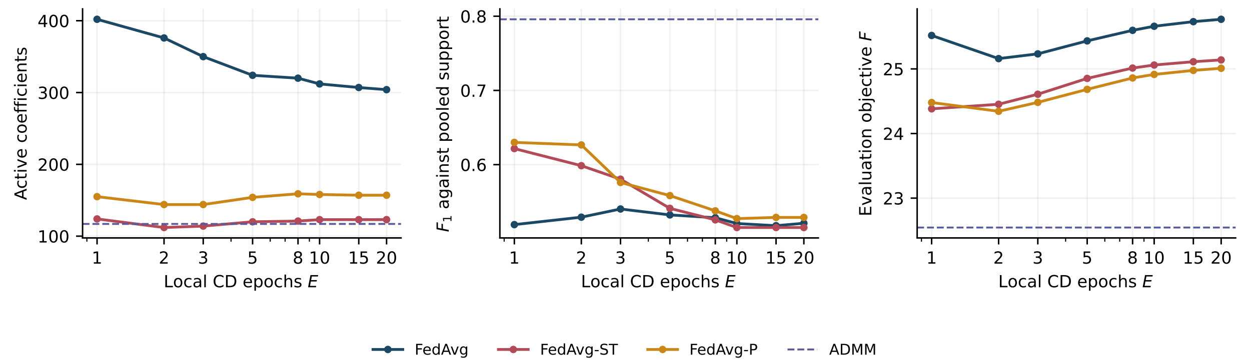


**Figure 4:** Fixed synthetic benchmark: active-set size (left), agreement with pooled support (middle), and evaluation objective (right). Dashed lines denote ADMM at $\bar{\lambda}$, solved to numerical tolerance. A flatter size curve does not imply more stable variable identities.

**Table 4:** Fixed synthetic benchmark ($n = 800$, $p = 600$, $K = 3$). The original generator and true coefficients were not supplied. Agreement is with the pooled minimum-MSE Lasso support (172 variables), not with ground truth. $F$ is the common evaluation objective. Selected epoch counts are shown; the complete grid and costs, including all five P candidates, are supplied in `code/real_data_results.csv`.

| Method | $E$ | Active | Precision | Recall | $F_1$ | $F$ |
|---|---|---|---|---|---|---|
| Pooled | — | 172 | 1.000 | 1.000 | 1.000 | 22.901 |
| Pooled-1SE | — | 98 | 0.969 | 0.552 | 0.704 | 22.604 |
| Local-only | — | 164 | 0.494 | 0.471 | 0.482 | 27.041 |
| Local-only-1SE | — | 83 | 0.639 | 0.308 | 0.416 | 26.015 |
| One-shot | — | 299 | 0.415 | 0.721 | 0.527 | 25.858 |
| ADMM | — | 117 | 0.983 | 0.669 | 0.796 | 22.545 |
| ADMM-CV | — | 172 | 1.000 | 1.000 | 1.000 | 22.909 |
| ADMM-CV-1SE | — | 96 | 0.969 | 0.541 | 0.694 | 22.618 |
| FedAvg | 1 | 402 | 0.371 | 0.866 | 0.519 | 25.518 |
| FedAvg-ST | 1 | 124 | 0.742 | 0.535 | 0.622 | 24.382 |
| FedAvg-ST-min | 1 | 402 | 0.371 | 0.866 | 0.519 | 25.518 |
| FedAvg-P | 1 | 155 | 0.665 | 0.599 | 0.630 | 24.480 |
| FedAvg | 2 | 376 | 0.386 | 0.843 | 0.529 | 25.160 |
| FedAvg-ST | 2 | 112 | 0.759 | 0.494 | 0.599 | 24.453 |
| FedAvg-ST-min | 2 | 376 | 0.386 | 0.843 | 0.529 | 25.160 |
| FedAvg-P | 2 | 144 | 0.688 | 0.576 | 0.627 | 24.344 |
| FedAvg | 3 | 350 | 0.403 | 0.820 | 0.540 | 25.233 |
| FedAvg-ST | 3 | 114 | 0.728 | 0.483 | 0.580 | 24.608 |
| FedAvg-ST-min | 3 | 350 | 0.403 | 0.820 | 0.540 | 25.233 |
| FedAvg-P | 3 | 144 | 0.632 | 0.529 | 0.576 | 24.482 |
| FedAvg | 20 | 304 | 0.408 | 0.721 | 0.521 | 25.768 |
| FedAvg-ST | 20 | 123 | 0.618 | 0.442 | 0.515 | 25.139 |
| FedAvg-ST-min | 20 | 304 | 0.408 | 0.721 | 0.521 | 25.768 |
| FedAvg-P | 20 | 157 | 0.554 | 0.506 | 0.529 | 25.010 |

At $E = 1$, FedAvg retains 402 predictors with agreement $F_1 = 0.519$, compared with 0.482 for the largest-site fit. ST retains 124 predictors with $F_1 = 0.622$; P retains 155 with $F_1 = 0.630$. ADMM at $\bar{\lambda}$ retains 117 and gives $F_1 = 0.796$. ADMM-CV under minimum-MSE selection has the same support as the pooled reference in this run, but slightly different coefficients and penalty. The pooled-objective identity guarantees equal solutions only at equal penalties. Neither support agreement nor its increase validates the unknown generating support. Pooled one-SE tuning retains

98 predictors, illustrating the dependence of the reference itself on tuning.

Thresholding reduces variation in active-set size across $E$, while ST's agreement declines from 0.622 at $E = 1$ to 0.515 at $E = 20$. Its $F_1$ range is larger than that of plain averaging. Thus a flatter size curve cannot be interpreted as stability of the selected variables, and this one benchmark cannot establish the cause.

ADMM attains $F = 22.545$, versus 25.518 for FedAvg at $E = 1$: a gap of 13.2% for the specified common objective. The FedAvg objective is smallest at $E = 2$ (25.160) and reaches 25.768 at $E = 20$. Meanwhile rounds fall from 28 to 6, upload from 393.8 to 84.4 KiB, and local CD work increases from 28 to 120 sweeps per site. These are separate optimisation and cost comparisons. P's released cost totals include all five threshold candidates; its selected-fit costs are also supplied. No distributed tuning cost is assigned to the centrally evaluated ADMM-CV paths.

# 9 Discussion

This study diagnoses how CD averaging, local epochs and penalty selection affect sparse estimation. The common-penalty comparison favours adapted FedDualAvg over CD averaging on both objective accuracy and support recovery at the reported selected-path budgets. Its additional step-size search must be counted when comparing complete procedures. When accurate minimisation of the pooled penalised objective is required, these results support an established composite method, with consensus ADMM supplying another numerically checked reference.

The effect of local epochs on selection within CD averaging depends on the aggregation procedure and penalty rule. The orthogonal calculation makes one distinction exact: local CD finishes in one sweep, so further local epochs have no effect even though averaging can enlarge the selected support. For general designs, an averaged fixed point need not minimise the pooled Lasso objective. Neither fact implies that every average is dense or that its statistical performance is determined by its optimisation gap alone.

The simulation comparisons support reporting tuning choices alongside aggregation choices. Local one-SE penalties improve plain averaging, while post-fit and per-round thresholds introduce additional selection steps. Their search spaces differ, and the resulting performance differences should not be assigned wholly to one mechanism. Likewise, favourable support recovery at an inherited penalty should not be interpreted as a universal advantage of a particular solver. ST remains useful as an inexpensive post-fit sparsification step for a CD workflow; P changes the iteration but requires five candidate fits and has no demonstrated advantage over FedDualAvg under matched total costs. Their role here is to quantify retrofit behaviour, including its prediction and computation costs.

Activity thresholds, numbers selected, and variable identities are distinct reporting choices. Sensitivity to the numerical activity threshold should be shown, and a nearly constant number of selected variables should not be called stable support without checking the identities. The fixed synthetic benchmark illustrates this distinction: thresholding makes the size curve flatter while its agreement with a pooled reference varies more with $E$. Where a generating support is unavailable, such agreement remains a descriptive comparison between estimators.

Communication accounting must include additional federated fits required for tuning. Post-fit thresholding is inexpensive under the stated upload convention; selecting a per-round threshold requires several complete trajectories. Local CD work and uploaded bytes can move in opposite directions as $E$ changes. Neither measure alone establishes a preferred schedule, especially when different algorithm families perform different local computations or use different stopping criteria.

**Limitations.** The exact support calculations require orthogonal local designs with $p \leq \min_j m_j$, outside the high-dimensional simulations. The general-design fixed-point result supplies no convergence rate or nonasymptotic bound linking $E$ to support error. The main study uses one sample size, one sparsity level, three sites, Gaussian designs and a shared coefficient vector. It does not establish uniform performance over other site counts, signal strengths, coefficient heterogeneity or response models. The observation-level one-SE rule is a tuning heuristic, especially with heterogeneous fixed sites; it is not uncertainty quantification for the selected support. A systematic comparison with stratified or nested tuning is outside this study.

The original fixed benchmark cannot validate recovery because its generating coefficients and generation procedure are unavailable. The article therefore contains no observational-data validation of high-dimensional support recovery or deployed multi-institutional application. The communication-budget comparison is a deterministic specialisation of FedDualAvg; it does not reproduce every stochastic setting or accelerated method in the composite federated literature. The common-penalty solver comparison and the locally tuned thresholding study use different search spaces; they do not establish a ranking of ST, P and FedDualAvg under matched tuning and total costs. Local computation also differs between the solver families.

Finally, keeping records at a site does not provide a formal privacy guarantee. Parameters and scalar summaries may reveal sample information (Zhu and Han, 2020). The experiments measure algorithmic and statistical behaviour, not differential privacy, secure aggregation, or operational network performance.

## Data and code availability

Code, recorded simulation seeds, raw and summary results, and the fixed synthetic benchmark are available at `https://github.com/KeivanBolouri/federated-lasso`. The accompanying source archive includes the scripts, fixed inputs, recorded results and orthogonal-design experiment needed to regenerate the numerical outputs, manuscript and Supplement S1. The original generator and true coefficients for the fixed synthetic benchmark were not supplied.

## Supplementary material

Supplement S1, *Supplementary numerical results*, contains the complete simulation grid in Table S1, the common total-upload-cap comparison in Table S2, and an orthogonal-design illustration in Table S3. It is supplied as a separate PDF with the manuscript source files.

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# A Additional numerical results

Table 5 varies the activity threshold in the fixed synthetic benchmark. All methods can retain exact zeros, and all can lose active coordinates when the threshold exceeds the magnitude of a nonzero coefficient. Thus threshold sensitivity must be described from the counts, not inferred solely from whether the estimator uses soft-thresholding.

**Table 5:** Active coefficients in the fixed synthetic benchmark at different activity thresholds. All methods may retain exact zeros; increasing the threshold also removes nonzero coefficients, including for ST and ADMM.

| Method | $E$ | $\epsilon = 10^{-6}$ | $10^{-4}$ | $10^{-2}$ | $10^{-1}$ |
|---|---|---|---|---|---|
| FedAvg | 1 | 402 | 402 | 357 | 103 |
| FedAvg-ST | 1 | 124 | 124 | 108 | 58 |
| FedAvg | 5 | 324 | 324 | 285 | 102 |
| FedAvg-ST | 5 | 121 | 120 | 107 | 54 |
| FedAvg | 20 | 304 | 304 | 271 | 107 |
| FedAvg-ST | 20 | 123 | 123 | 113 | 59 |
| ADMM | — | 117 | 117 | 110 | 65 |

The complete simulation grid is provided in Supplementary Table S1 and in the raw and summary CSV files. These include estimation and prediction errors, support metrics, convergence diagnostics, and selected-fit and full-selection costs.